Fair District Maps Through Smart Computer Algorithms

Edge-Weighted Recursive Bisection Explained

Giovanni Della-Libera

January 2026

What We'll Cover

1. The Redistricting Problem
2. Understanding Graphs
3. Recursive Bisection Algorithm
4. The Key Innovation: Edge Weights
5. Real Examples
6. National Results
7. Conclusion

Why Do We Draw Congressional Districts?

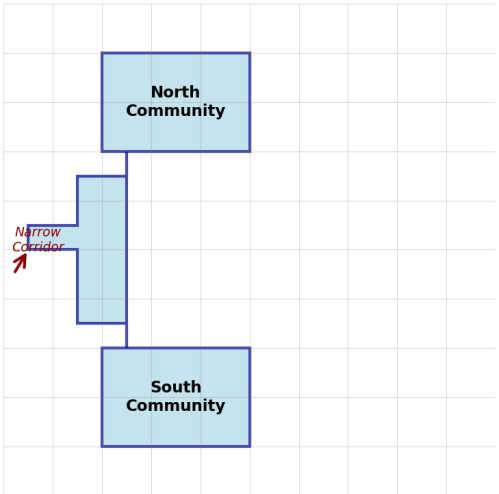
The Goal:

- Every 10 years, new census
- States get a certain number of representatives
- Need to draw district boundaries
- Each district $\approx$ 760,000 people (2020)

The Challenge:

- Millions of possible ways to draw lines
- Politicians often draw maps to favor their party
- This is called **gerrymandering**

Example of Gerrymandering:
Connecting Distant Communities via Narrow Corridor



Inspired by Illinois 4th District ("The Earmuffs")

What Makes a "Good" District Map?

Constitutional Requirements:

1. **Equal Population**: Each district must have approximately the same number of people
2. **Contiguous**: Districts must be connected (you can walk from any part to any other part)

Desirable Properties:

- **Compact**: Districts should be reasonably circular/square, not weird snaking shapes
- **Fair**: Maps shouldn't systematically favor one political party
- **Transparent**: The process should be understandable and auditable

Can we use math and computers to create better maps?

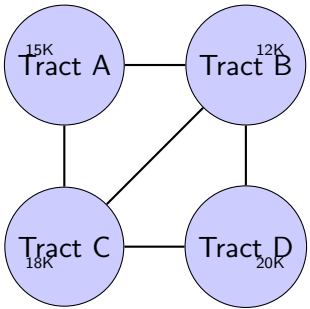
What's a Graph? (Not the x-y kind!)

A **graph** is a way to represent connections:

- **Vertices** (nodes): The things we're connecting
- **Edges**: The connections between them

For Redistricting:

- Each **census tract** is a vertex
- Census tracts that touch each other have an **edge**
- Each vertex has a **weight** (population)



Key Insight: Redistricting is a *graph partitioning* problem—we need to split the graph into groups (districts) with equal total weight (population).

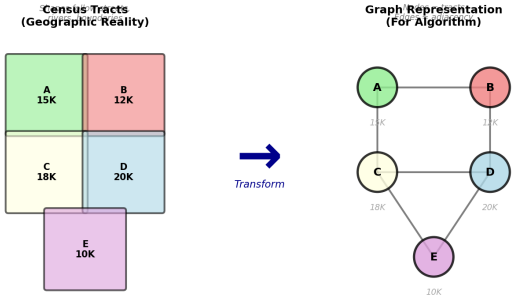
From Real Geography to Graph

Actual Census Tracts:

- Small geographic areas
- Typically 1,200-8,000 people
- About 85,000 tracts nationwide
- Shapes follow streets, rivers, etc.

Why Use Graphs?

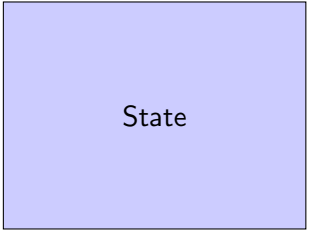
- Computers are great at graph problems
- Preserves all geographic relationships
- Automatically ensures contiguity



The Basic Idea: Split in Half, Repeatedly

Recursive Bisection: A "divide and conquer" approach

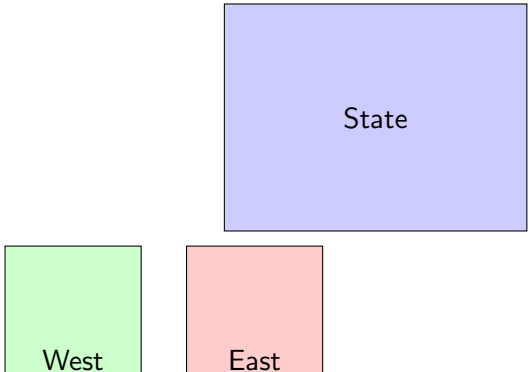
1. Start with the whole state
2. **Split it in half** (balanced by population)
3. Split each half in half
4. Keep going until you have the right number of districts



The Basic Idea: Split in Half, Repeatedly

Recursive Bisection: A "divide and conquer" approach

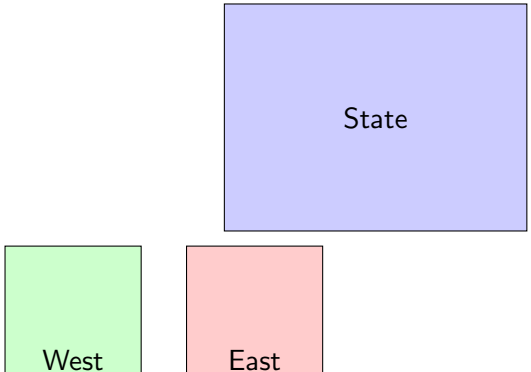
1. Start with the whole state
2. **Split it in half** (balanced by population)
3. Split each half in half
4. Keep going until you have the right number of districts



The Basic Idea: Split in Half, Repeatedly

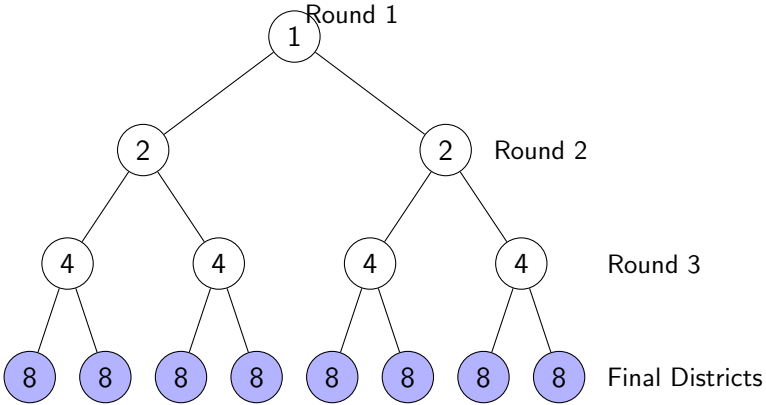
Recursive Bisection: A "divide and conquer" approach

1. Start with the whole state
2. **Split it in half** (balanced by population)
3. Split each half in half
4. Keep going until you have the right number of districts



Example: 8 Districts = 3 Rounds of Splitting

Round 1: $1 \rightarrow 2$ regions
Round 2: $2 \rightarrow 4$ regions
Round 3: $4 \rightarrow 8$ districts



How Do We Actually Split?

We use a tool called METIS:

- Specialized graph partitioning software
- Developed at University of Minnesota
- Used in engineering, scientific computing, etc.
- Very good at splitting graphs efficiently

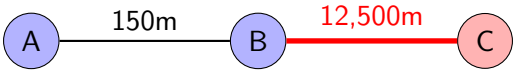
What METIS does at each split:

1. Takes the current region (as a graph)
2. Finds a way to divide it into two parts
3. Makes sure each part has the right population
4. Makes sure each part is connected
5. Tries to minimize something (more on this next!)

Baseline Approach: Treating All Connections Equally

Unweighted Recursive Bisection:

- All edges are treated the same
- METIS minimizes the **number of edges cut**
- Doesn't care about boundary *length*



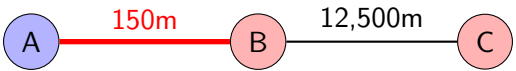
Cut costs: 1 edge

Problem: Cutting between B and C adds 12.5 km of boundary!

Our Innovation: Edge Weights Based on Real Length

Edge-Weighted Recursive Bisection:

- Each edge has a **weight** = actual boundary length
- METIS minimizes the **total weight of edges cut**
- Directly minimizes total perimeter!



Cut costs: 150 (much better!)

Insight: METIS now strongly prefers to cut short boundaries, not long ones.

Why Does This Matter?

Compact districts have shorter perimeters!

Circle: Most compact shape

- Perimeter $= 2\pi r \approx 6.28r$
- Polsby-Popper score $= 1.0$

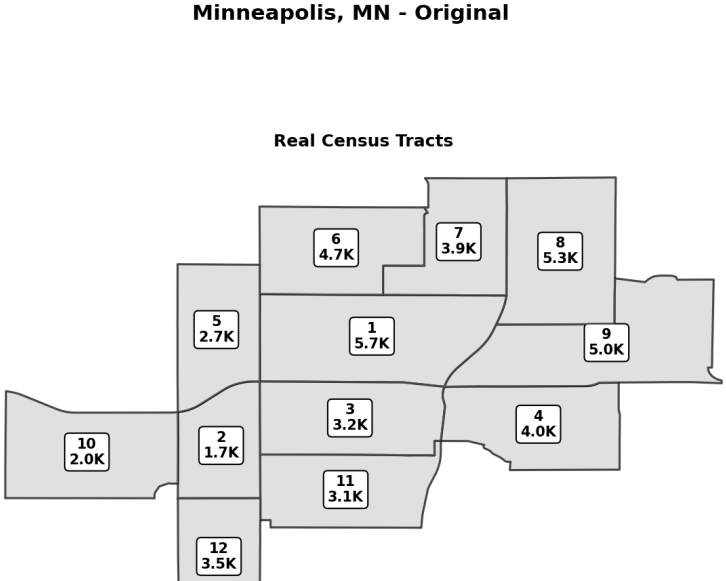
Squiggly shape: Less compact

- Same area, much longer perimeter
- Lower Polsby-Popper score

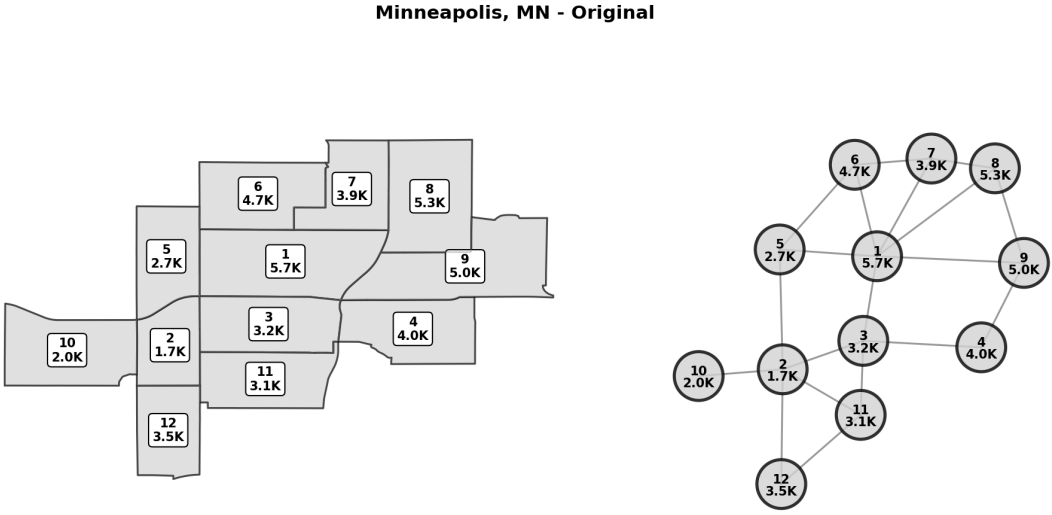
By minimizing total perimeter, we automatically get more compact districts!

$$\textit{Polsby-Popper formula: } PP = \frac{4\pi \times \text{Area}}{\text{Perimeter}^2}$$

Real Geography: Minneapolis Census Tracts

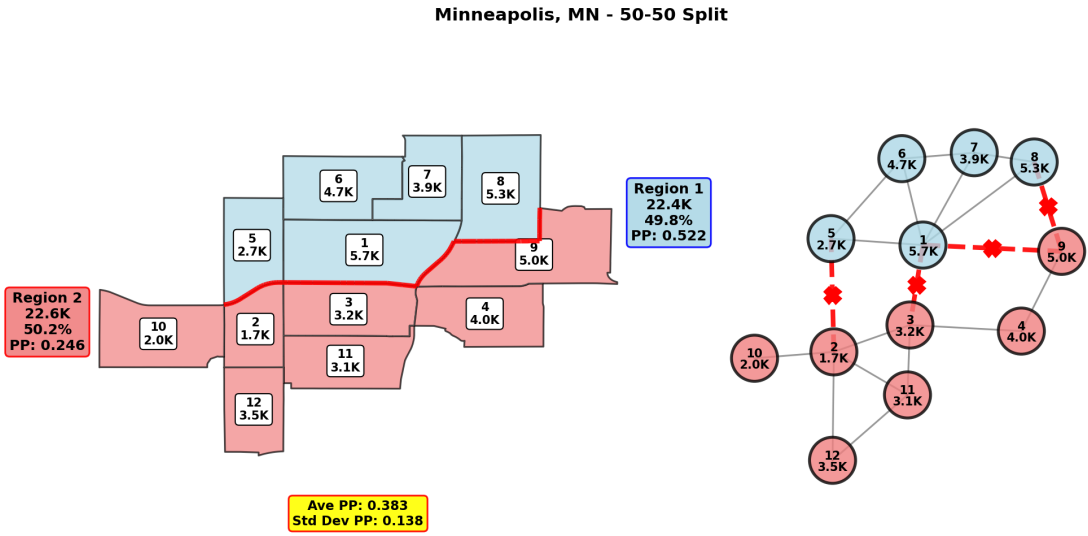


From Real Geography to Abstract Graph



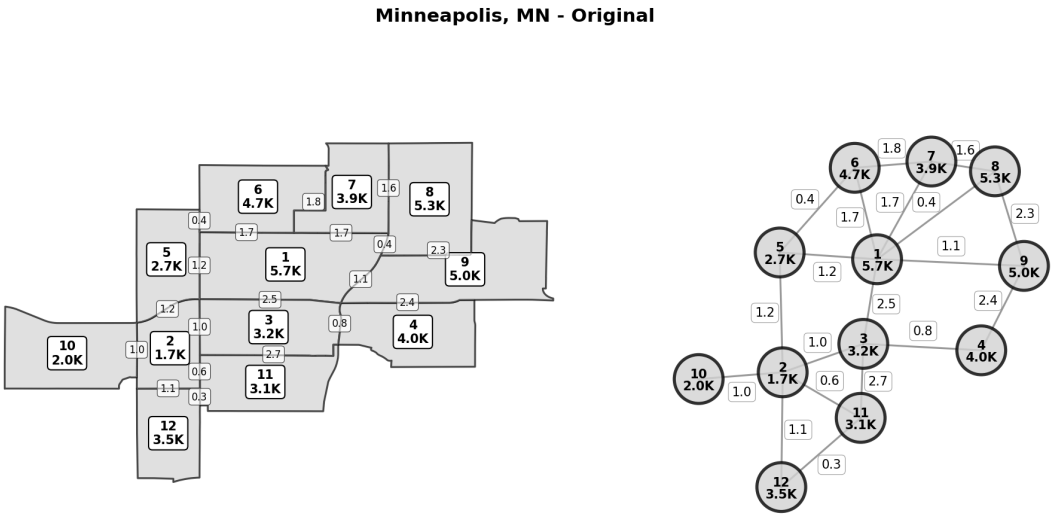
Left: Actual census tracts from Minneapolis area with irregular shapes

Unweighted Partition: Why It Fails



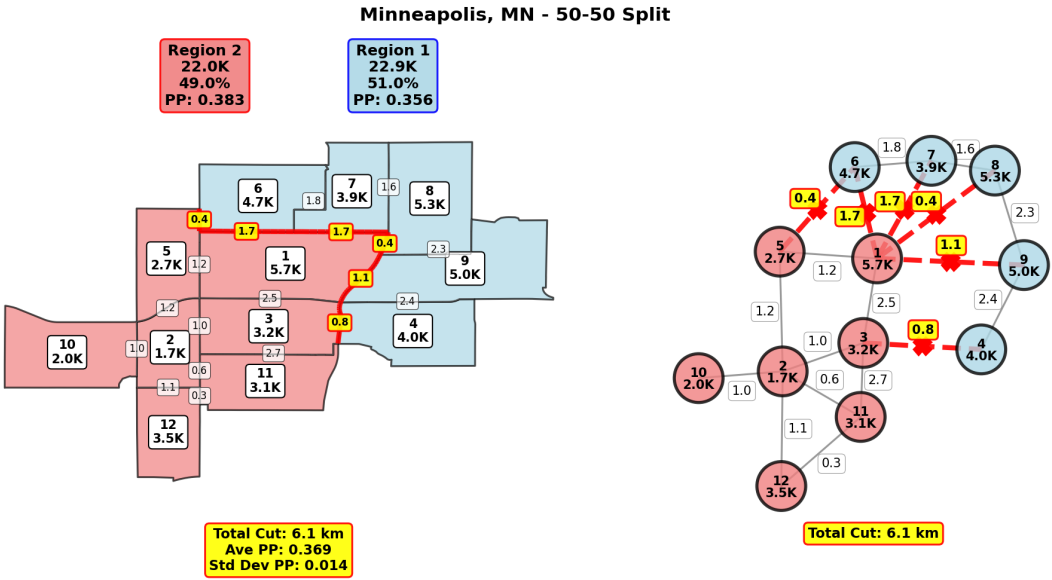
Traditional approach treats all edges equally. Result: Cuts 23.6 km of boundaries!

Edge Weights: Show All Boundaries

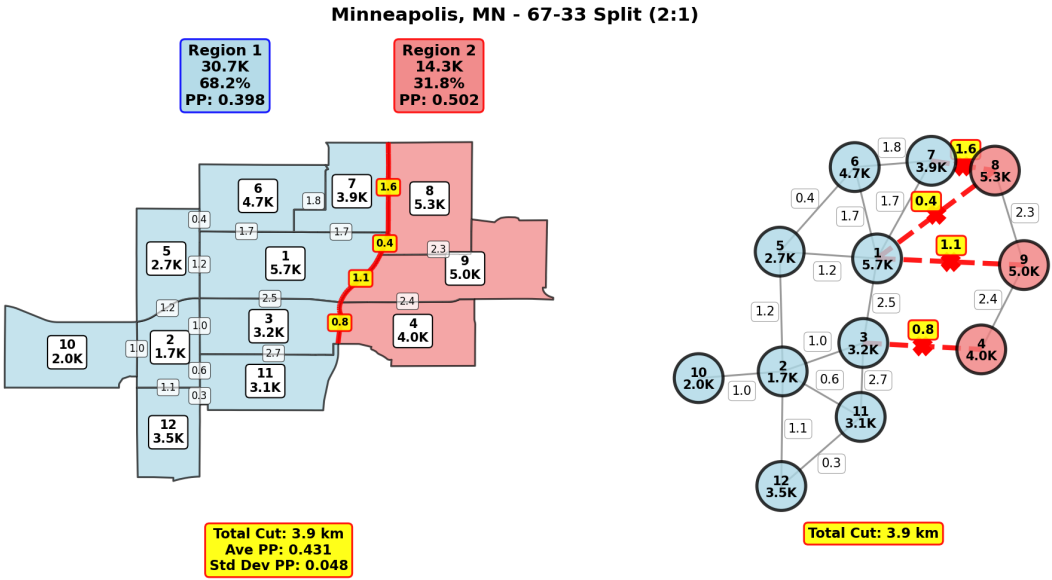


Now we see all edge weights (boundary lengths in km). Longer boundaries should be more expensive to

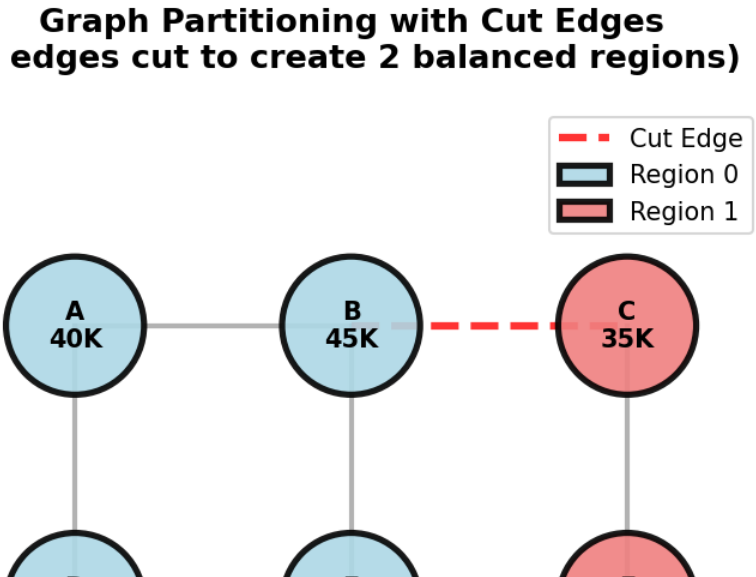
Edge-Weighted Partition: Much Better!



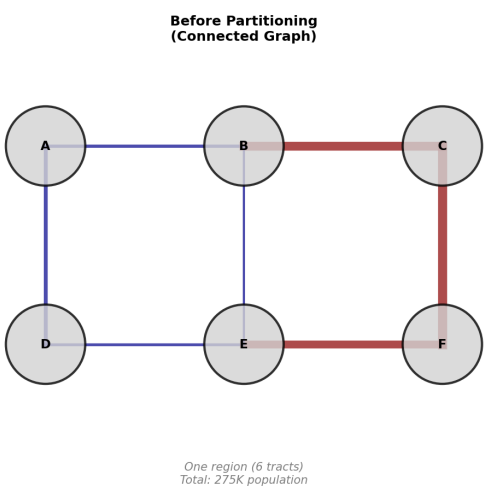
Works for Unequal Splits Too



The Cut Operation (Schematic)



Before and After Partitioning



Result: Two balanced districts separated by cutting only the shortest boundaries.

How We Call METIS

METIS Graph File Format:

```
6 7 011 2                % 6 vertices, 7 edges, format 011
2 5.2 1                  % Tract A: neighbor 2, weight 5.2
1 5.2 3 12.8 2           % Tract B: A (5.2km), C (12.8km)
2 12.8 3                 % Tract C: B (12.8km)
...
```

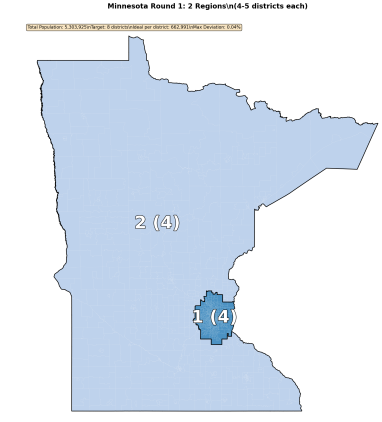
Calling METIS:

```
gpmetis graph.txt 2
```

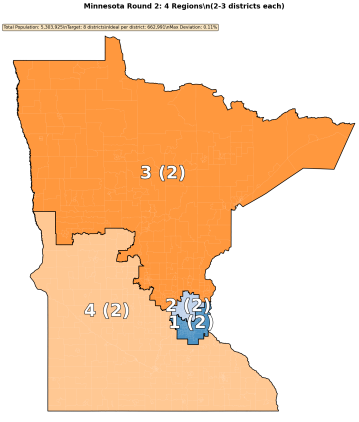
Output: One partition ID per tract (0 or 1)

```
0  (Tract A -> District 0)
0  (Tract B -> District 0)
1  (Tract C -> District 1)
...
```

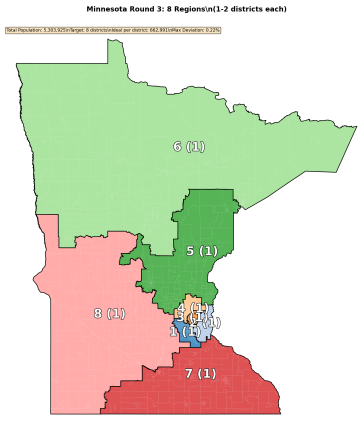
Minnesota: 8 Districts (Perfect Power of 2)



Round 1: 2 regions



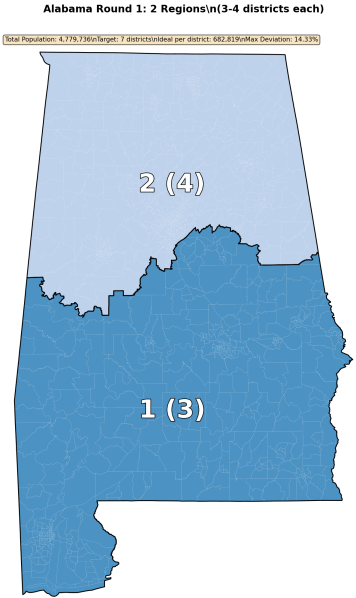
Round 2: 4 regions

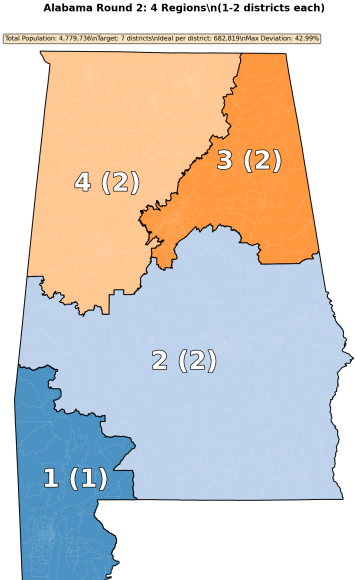


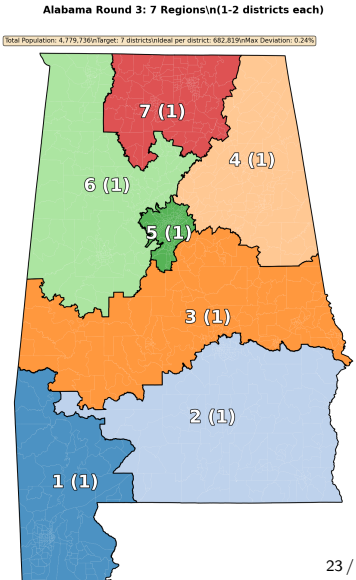
Round 3: 8 districts

Notice: Each split creates two equal-sized regions. Final districts are compact with smooth boundaries.

Alabama: 7 Districts (Not a Power of 2)







Comparing Approaches: Minnesota Results

Method	Mean Polsby-Popper	vs Baseline
Baseline (Unweighted)	0.288	—
Edge-Weighted	**0.468**	+62%
Enacted 2020 Map	0.287	+0%

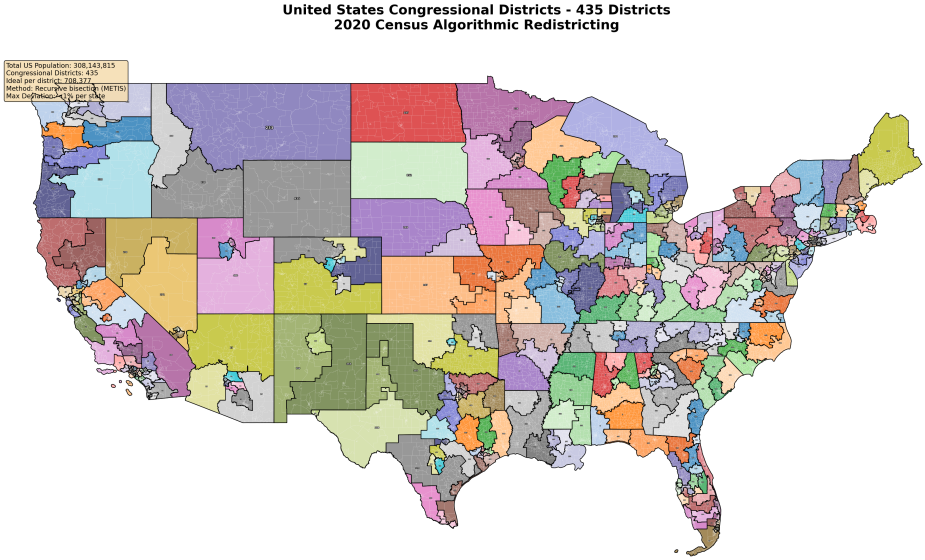
Takeaway: Edge weights make a huge difference! Our algorithm produces more compact districts than both baseline and the actual enacted map.

Comparing Approaches: Alabama Results

Method	Mean Polsby-Popper	vs Baseline
Baseline (Unweighted)	0.246	—
Edge-Weighted	**0.369**	+50%
Enacted 2020 Map	0.297	+21%

Takeaway: Even with 7 districts (odd number), edge-weighted bisection produces compact districts that beat the enacted map!

National Map: All 435 Districts



Results Across All 50 States

All 435 Congressional Districts (2020 Census)

Method	Mean PP	Improvement
Baseline Recursive Bisection	0.235	—
Edge-Weighted Bisection	**0.367**	+56%
Enacted 2020 Districts	0.305	+30%

Key Results:

- **37 of 50 states**: Our algorithm beats enacted maps
- **Partisan-drawn states**: Biggest improvements
  - Illinois: +174%
  - Louisiana: +104%
  - New Hampshire: +102%

Why This Matters

Advantages of Algorithmic Redistricting:

1. **Objectivity**: No human bias or political manipulation
2. **Transparency**: Anyone can run the algorithm and verify results
3. **Reproducibility**: Same inputs always produce same outputs
4. **Compactness**: Automatically produces compact districts
5. **Speed**: Runs in minutes, not months

Limitations:

- Doesn't consider communities of interest
- Doesn't try to create competitive districts
- Pure geography-based approach

Use Case: Provides a *neutral baseline* to measure gerrymandering

Summary

What We Learned:

1. Redistricting can be viewed as a **graph partitioning problem**
2. **Recursive bisection**: Split in half repeatedly until we have enough districts
3. **Edge weights**: Using real boundary lengths dramatically improves compactness
4. **METIS**: Powerful tool that does the hard work of finding good partitions
5. Works for both even (Minnesota, 8) and odd (Alabama, 7) district counts

The Big Idea:
By encoding geographic information (boundary lengths) into the graph, we can automatically create compact, fair districts without any human intervention.

Questions?

Thank You!

For more details:

- Full paper: `papers/03_combined_recursive_bisection/`
- Code: `github.com/yourusername/redistricting`
- Interactive maps: Visit our web dashboard